Exact lattice gauge theory on the octahedron

An illustrated guide

physics, figures, and code verification

Manuscript:	*Exact lattice gauge theory on the octahedron*
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Target journal:	Nuclear Physics B
Version:	2026-10-24 (revision v3)
Code:	`https://github.com/csoftxyz/RIA_EISA/tree/main/08/code_and_logs`

Every number in this guide is recomputed by the companion scripts and asserted in `verify_all.py`. Logs are in `logs/`; the full test report is `TEST_REPORT.md`.

1 Five sentences

If you read only five sentences

1. On the octahedron (V=6, E=12, F=8), pure compact $U(1)$ lattice gauge theory is exactly solvable, $Z_F(\beta) = \sum_n I_n(\beta)^F$, and the result depends on the number of faces alone—not on the genus.
2. On a closed surface the Wilson loop obeys an exact area law, $\langle W \rangle = \sum_n I_n^A I_{n+1}^{F-A} / \sum_n I_n^F$: it depends on the enclosed area A, not on the shape of the loop.
3. The finite-volume string tension is 0.69% low, and the correction is not a power of $1/F$ but exponential, $\sigma_\infty - \sigma_F = 3.5565\, e^{-0.80656F}[1 + O(r^F)]$, with an analytically derived prefactor and a controlled remainder.
4. The Villain self-dual coupling is extensive, $\beta_* = F/2\pi$ ($4/\pi$ for the octahedron), so $\beta = 1$ is self-dual on no lattice; the three natural Wilson–Villain dictionaries also miss $4/\pi$.
5. Within this minimal construction, chiral fermions and the standard-model gauge algebra do not arise. This is not a cosmological statement: it is a set of five checkable criteria, scoped to this graded space and these bracket conventions.

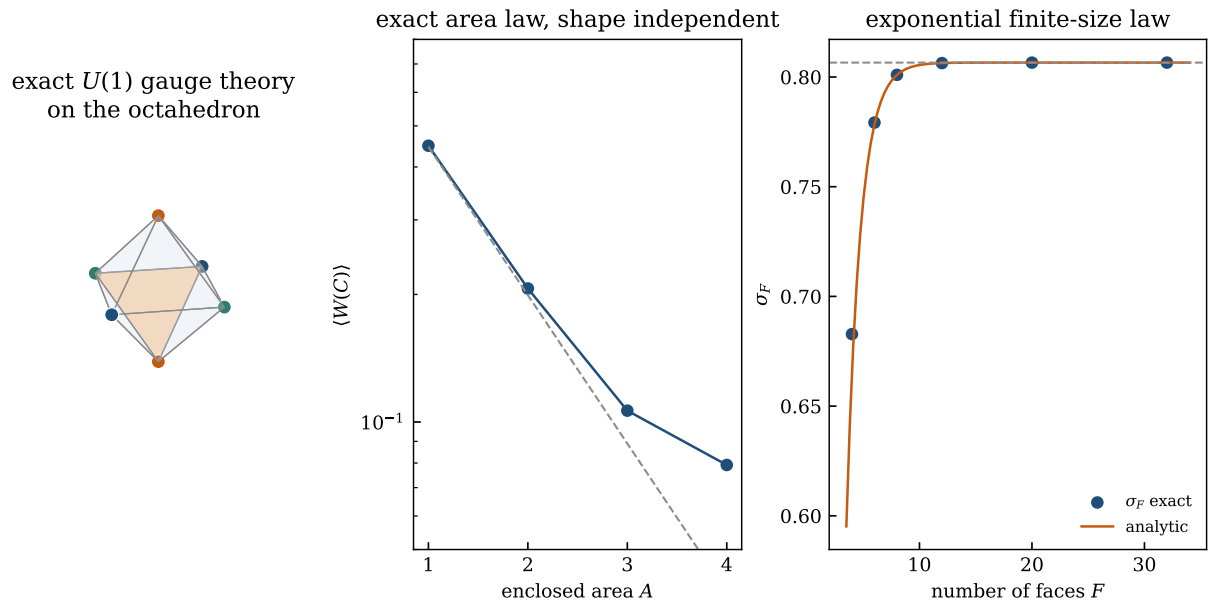


Figure 1: At a glance: the carrier (octahedron, V=6, E=12, F=8); the exact area law (only the enclosed area matters); and the finite-volume tension with its analytic exponential law. Every curve is regenerated by the companion code.

The guide follows the logical order: geometry → exact solution → area law → finite size → thermodynamics → duality → fermions → algebra. Derivations are in the paper and its appendices; here the emphasis is on the physical picture and on the evidence chain.

2 Why the octahedron: geometry and Gauss law

The octahedron is the six-vertex triangulation of S^2, and it is unique: $E = 3V - 6 = 12$ forces degree 4; the complement of the graph in K_6 is a perfect matching; all perfect matchings are equivalent under relabelling, so the graph is $K_{2,2,2}$; it has exactly eight triangles, each edge in exactly two, vertex links are 4-cycles, $\chi = 2$; a 3-connected planar graph has a unique embedding (Whitney). The exhaustive enumeration is in `triangulation_uniqueness.py`.

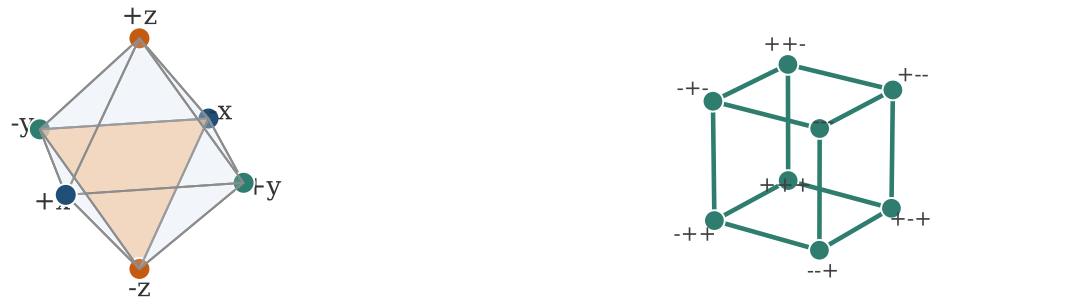


Figure 2: Left: the octahedron $K_{2,2,2}$, with vertices the two signs of three coordinate axes and faces the eight sign triples; vertices are coloured by axis. Right: the dual graph is the cube, one vertex per face, adjacent when the faces share an edge. Connectedness of the dual graph is the geometric reason why all face fluxes are equal.

Expand each face weight in characters, $e^{\beta\cos\theta_f} = \sum_{n_f} I_{n_f}(\beta)e^{in_f\theta_f}$, and integrate each link: the result is $\delta_{n_f n_{f'}}$. Neighbouring faces must carry the same integer flux; the dual graph is connected, so all faces share one integer n:

$$Z_F(\beta) = \sum_{n\in\mathbb{Z}} \big[I_n(\beta)\big]^F. \tag{1}$$

This single step gives analyticity (positive series), F**-universality** (dependence on the number of faces only) and the fact that there is exactly one parameter, β. On the sphere there are no non-contractible loops, no flat-connection moduli, and $\sum_f \theta_f \equiv 0 \pmod{2\pi}$ makes a θ term vanish identically. This is what we call **topological rigidity**: the gauge content is fixed by the Gauss law, not by thermodynamics.

"Finite volume is analytic" is not the point

Of course a finite system is analytic. What has content is that the *infinite-volume* theory with the same action also has no transition (two-dimensional compact $U(1)$ confines at every β), while the finite-volume deviation is exponentially small and computable in closed form. The octahedron is therefore not a toy: it is a controlled window onto infinite-volume physics.

3 Exact area law: area, not shape

A Wilson loop inserts one unit of flux on every link it crosses, so the equality of neighbouring face charges is shifted by one across the loop: the two sides carry n and $n+1$. If the loop separates the faces into A and $F-A$,

$$\langle W(C)\rangle = \frac{\sum_n I_n(\beta)^A I_{n+1}(\beta)^{F-A}}{\sum_n I_n(\beta)^F}. \tag{2}$$

It depends only on A, not on the length or shape of the loop—all loops on the sphere are contractible. At $\beta = 1$, $F = 8$, the values for $A = 1, 2, 3, 4$ are $0.4488851,\ 0.2066158,\ 0.1063616,\ 0.0791731$.

The large-F limit has a transparent saddle-point reading: the dominant term is $n = -1$ (making $I_{n+1} = I_0$), giving r^A. At the symmetric split $A = F/2$ the two flux sheets (one on either side) are degenerate, so $\langle W\rangle = 2r^{F/2}$; the factor two counts screening channels, not a fit parameter.

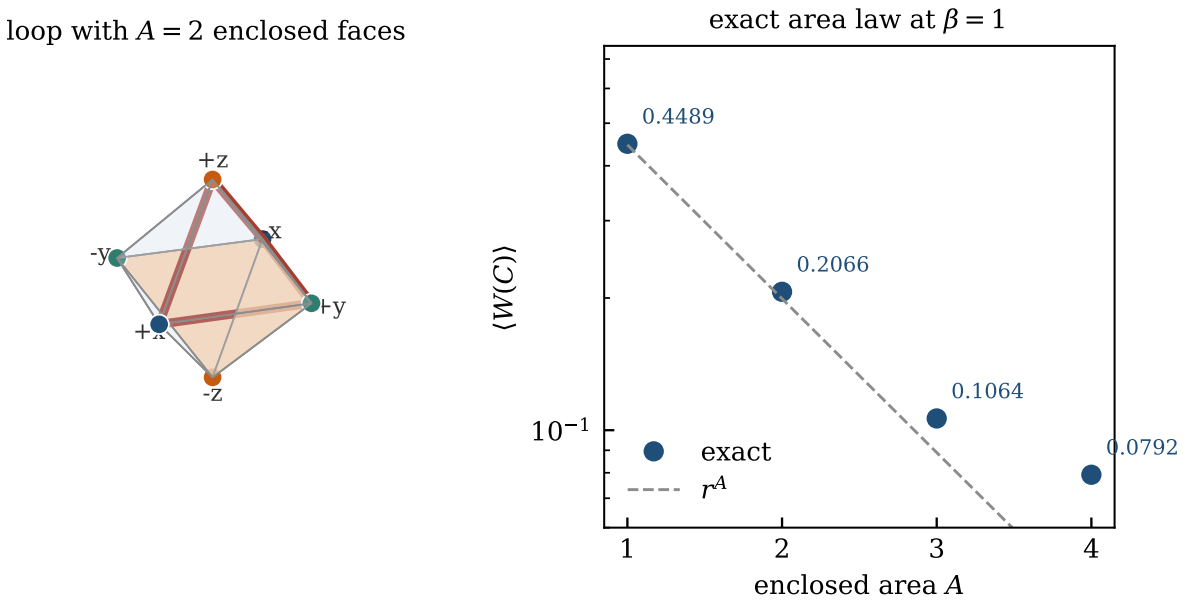


Figure 3: Left: a four-edge loop (red) enclosing two faces (shaded). Right: the exact $\langle W_A \rangle$ at $\beta = 1$ compared with r^A (dashed, $r = I_1/I_0$). At $A = 4$ the two screening channels are degenerate and a factor two appears.

4 Finite-size correction: exponential, derived, bounded

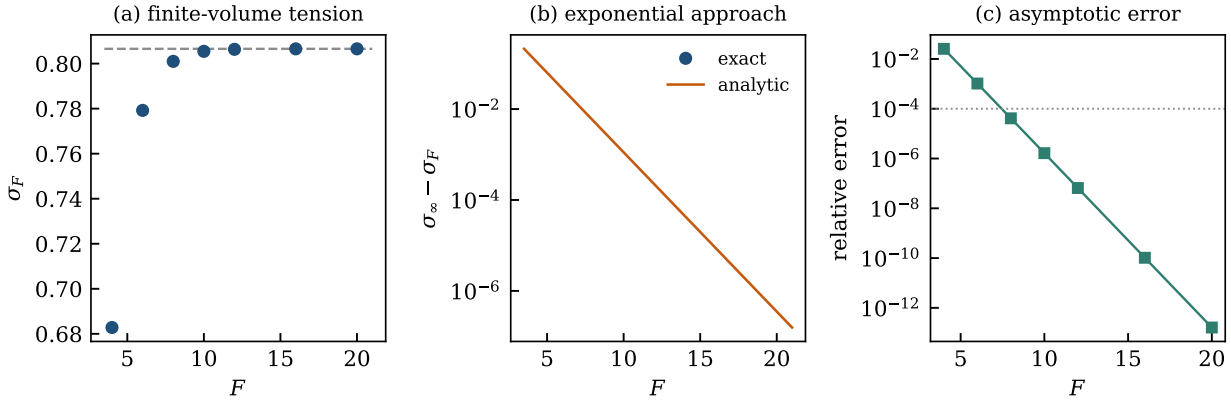


Figure 4: (a) Exact σ_F versus σ_∞. (b) The deficit on a logarithmic scale compared with the analytic formula $r^{F-2}[1 + (I_2/I_1)r - 2r^2]$. (c) Relative error of the asymptotic formula: 4×10^{-5} at $F = 8$ and 6×10^{-8} at $F = 12$.

Write $x = r^{F-2}$ and $A_1 = 1 + (I_2/I_1)r$. Keeping the exact denominator factor,

$$\frac{\langle W_1 \rangle}{r} = \frac{1 + xA_1 + O\big((I_2/I_0)^{F-1}\big)}{1 + 2r^2 x + O\big((I_2/I_0)^F\big)} = 1 + x\big(A_1 - 2r^2\big) + x\,O(r^F), \tag{3}$$

and for $A < F/2$, with $x_A = r^{F-2A}$,

$$\frac{\langle W_A \rangle}{r^A} = 1 + x_A\big(A_A - 2r^{2A}\big) + x_A\,O(r^F). \tag{4}$$

The remainder is bounded explicitly:

$$\frac{|\text{remainder of } \langle W_1 \rangle / r\,|}{x} \le 2r^F + O(r^{2F}) + O\left(\frac{(I_2/I_0)^{F-1}}{x}\right). \tag{5}$$

At $\beta = 1$, $F = 8$ this bound is 3.2×10^{-3} against a measured 2.2×10^{-3}; at $F = 12$, 1.3×10^{-4} against 1.2×10^{-4}. Hence

$$\sigma_\infty - \sigma_F = r^{F-2}\Big[1 + \frac{I_2}{I_1} r - 2r^2\Big]\big[1 + O(r^F)\big] = 3.5565\, e^{-0.80656\, F}\big[1 + O(r^F)\big]. \tag{6}$$

The prefactor $3.5565 = e^{2\sigma_\infty}[1 + (I_2/I_1)r - 2r^2]$ is analytic.

Why this is stronger than "the numbers agree"

The table in the paper lists exact value, asymptotic value, relative error, and the ratio $R_F = (\langle W\rangle/r - 1)/r^{F-2}$, which converges to the analytic bracket 0.708692… from below. The range of validity is explicit: $A < F/2$ and $r^{F-2A} \ll 1$; at $\beta = 1$ this means 1% at $F \gtrsim 6$ and 10^{-4} at $F \geq 8$. The $F = 4$ row is 2.6% low because the bound is not yet small there. This is derivation followed by verification, not curve fitting.

5 Thermodynamics: a crossover, not a transition

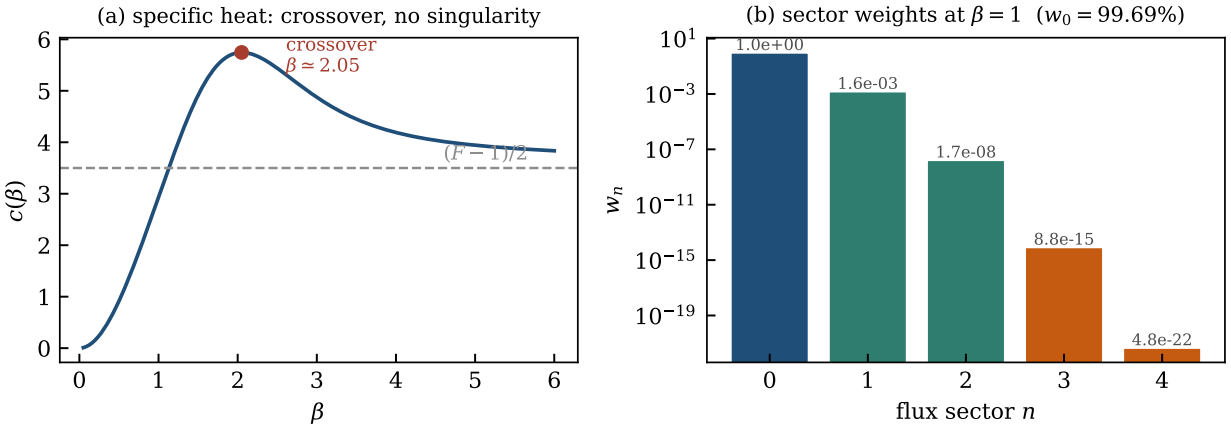


Figure 5: (a) Specific heat: a broad hump 5.745 at $\beta \simeq 2.05$, approaching the free-field value $(F-1)/2 = 3.5$; analytic everywhere. (b) Flux-sector weights at $\beta = 1$: $w_0 = 99.69\%$, with the $\mathbb{Z}_3$ charge almost entirely in $n \equiv 0$.

$c(\beta) = \beta^2 \operatorname{Var}(\sum_f \cos\theta_f)$, equal to 2.932656 at $\beta = 1$. The peak position can be computed exactly and is not tied to any non-analyticity, order parameter, or change of the flux distribution. A specific-heat hump on a closed surface is not evidence for a transition unless the infinite-volume limit of the same action is separately shown to be singular—which, for two-dimensional compact $U(1)$, it is not.

6 Duality: $\beta_* = F/2\pi$, and why it cannot pick a coupling

After Poisson resummation and the link integrals, the Villain model reduces to

$$Z_V(\beta) = (2\pi\beta)^{-F/2} \sum_{m\in\mathbb{Z}} e^{-Fm^2/(2\beta)} = (2\pi\beta)^{-F/2}\theta_3\Big(0, e^{-F/(2\beta)}\Big). \tag{7}$$

The modular transformation gives an exact duality $\beta \to \beta_*^2/\beta$ with

$$\beta_* = \frac{F}{2\pi}. \tag{8}$$

This is a key correction: the self-dual coupling is **extensive**, because β multiplies an F-fold sum. For the octahedron $\beta_* = 4/\pi = 1.2732$; for the cube ($F = 6$) it is 0.9549; $\beta_* = 1$ would require $F = 2\pi$. Hence "$\beta = 1$ is the self-dual point" is a convention, not a property of the model.

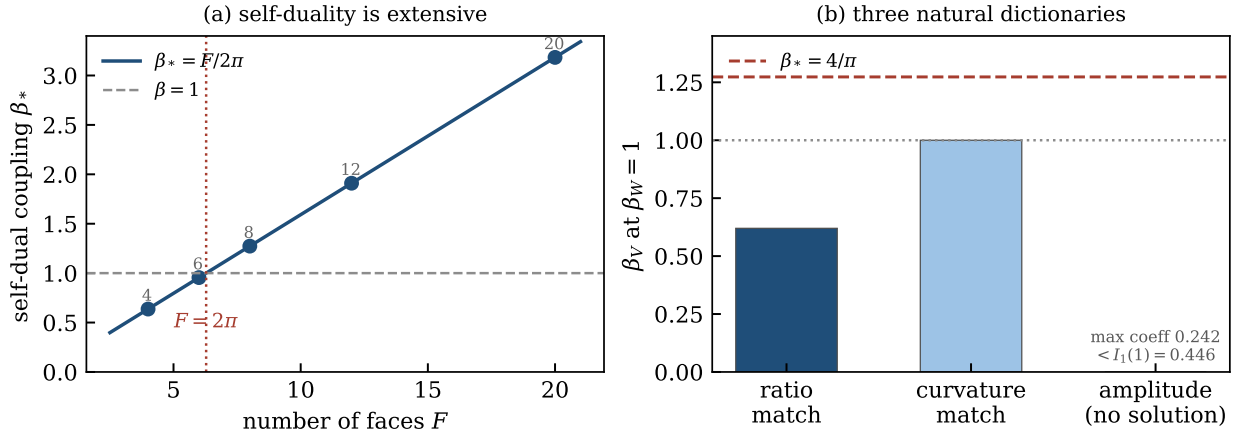


Figure 6: (a) The Villain self-dual coupling grows linearly with the number of faces; $\beta = 1$ crosses it only at $F = 2\pi$, which is not an integer. (b) The three natural Wilson–Villain dictionaries at $\beta_W = 1$; none reaches $4/\pi$.

The three dictionaries: ratio matching gives $\beta_V = 0.6199$; small-angle (curvature) matching gives 1; the first Fourier coefficient has maximum $0.24197 < I_1(1) = 0.44639$, so there is no solution. Conversely, the Wilson coupling that ratio-matches onto $4/\pi$ is $\beta_W = 1.8700$. The duality identity is verified numerically to 10^{-40}.

7 Fermions: massless versus full operator

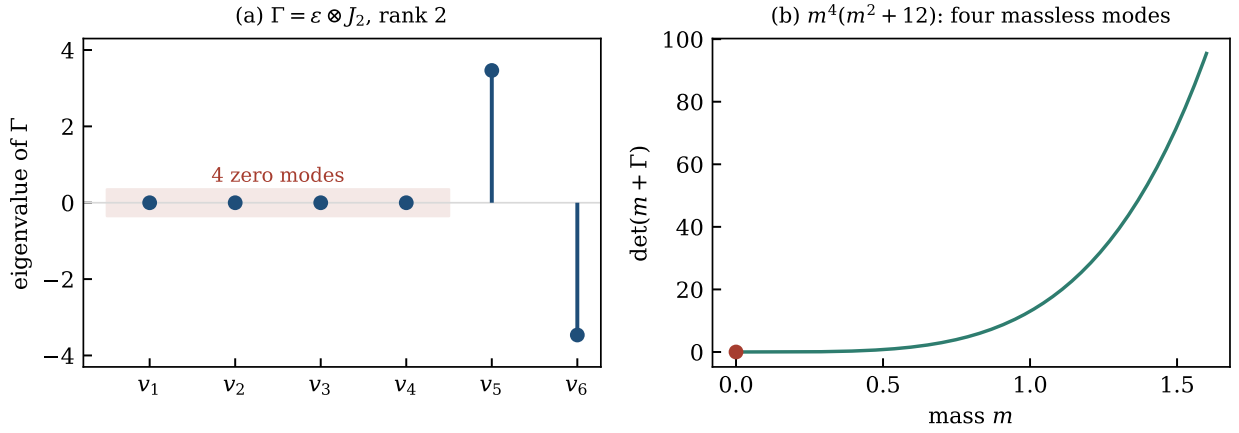


Figure 7: (a) Spectrum of $\Gamma = \varepsilon \otimes J_2$: four zeros and a pair $\pm 2\sqrt{3}i$; $\operatorname{rank}\Gamma = 2$. (b) At the flat connection, $\det(m + \Gamma) = m^4(m^2 + 12)$: a fourfold zero at $m = 0$.

We distinguish two operators from the start:

$$D_0 = t\,\Gamma \circ U \quad \text{(massless)}, \qquad D(m) = m\not{K} + D_0 \quad \text{(full)}, \tag{9}$$

with $U_{ij} = e^{i\theta_{ij}}$, $\theta_{ji} = -\theta_{ij}$, and $\circ$ the elementwise product (not the matrix product: $\det D_0 \neq \det\Gamma \det U$). Because $\Gamma_{ji} = -\Gamma_{ij}$,

$$(D_0)^\dagger_{ij} = \overline{\Gamma_{ji} U_{ji}} = -\Gamma_{ij} e^{i\theta_{ij}} = -(D_0)_{ij}, \tag{10}$$

so $D_0^\dagger = -D_0$ and $D(m)^\dagger = m\not{K} - D_0$; hence $\det(m + D_0)^* = \det(m - D_0)$. For $m = 0$, D_0 is anti-Hermitian of even dimension, so $\det D_0 \in \mathbb{R}$ for every configuration—real but sign-indefinite.

For $m > 0$, $\det D(m)$ is generically complex. In both cases the minimal positive analytic weight is the doubled one,

$$W_m = \det D(m) \det D(-m) = |\det D(m)|^2 \geq 0, \tag{11}$$

which at $m = 0$ is simply $(\det D_0)^2$.

Operator dictionary

object	definition	properties
D_0	$t\,\Gamma \circ U$	$D_0^\dagger = -D_0$; $\det D_0 \in \mathbb{R}$; $\det D_0 = 0$ at $U = \mathbb{1}$
$D(m)$	$m\mathbb{1} + D_0$	$D(m)^\dagger = m\mathbb{1} - D_0$; $\det(m + D_0)^* = \det(m - D_0)$
W_m	$\det D(m) \det D(-m)$	$= \|\det D(m)\|^2 \geq 0$

Chirality is stated precisely as *absence of a sublattice grading*: there is no vertex-diagonal $X = \mathrm{diag}(\pm 1)$ with $\{X, \Gamma\} = 0$. Such a grading is a two-colouring, and the octahedron has triangles. This does not exclude a non-local operator anticommuting with Γ; it excludes the sublattice symmetry a lattice chiral fermion would need.

At the flat connection $U = \mathbb{1}$, $D_0 = \Gamma$ has four exact zero modes,

$$\ker \Gamma = \mathrm{span}\{e_a^+ - e_a^- \ (a = x, y, z),\ \textstyle\sum_a e_a^+\}, \tag{12}$$

while the full operator has $\det(m\mathbb{1} + \Gamma) = m^4(m^2 + 12)$; for $m \neq 0$ these four modes have eigenvalue m and are not zero modes of $D(m)$. They are not protected across the ensemble: for random configurations the spectrum of D_0 is gapped, with $\min|\lambda|$ median 0.334, fifth percentile 0.032, and no configuration below 10^{-3} in a sample of 400. Note that $\det D_0$ is not identically zero (e.g. 8.375 for a random configuration): the vanishing at $U = \mathbb{1}$ is exactly the zero-mode condition.

8 The algebraic layer, with its scope

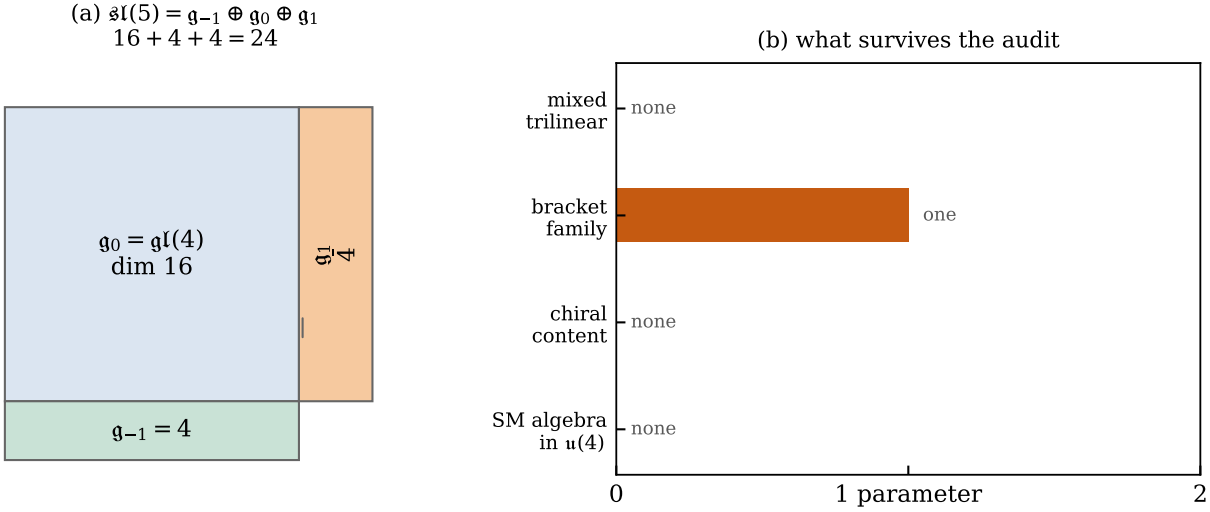


Figure 8: (a) The parabolic grading of $\mathfrak{sl}(5)$: a 4×4 $\mathfrak{gl}(4)$ block (16) plus two four-dimensional modules, $16 + 4 + 4 = 24$. (b) Audit outcome: no mixed trilinear; the mixed bracket is a one-parameter family that vanishes for an $\mathfrak{sl}(4)$ bracket; neither chiral content nor the standard-model algebra is present.

Identify $\mathfrak{u}(4) \oplus 4 \oplus \bar{4}$ with the parabolic grading

$$\mathfrak{sl}(5) = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1, \qquad \mathfrak{g}_0 \cong \mathfrak{gl}(4), \quad \mathfrak{g}_{-1} = 4,\ \mathfrak{g}_1 = \bar{4}. \tag{13}$$

The $\mathfrak{u}(4)$-invariant tensor census gives one bilinear and no trilinear to $\mathbb{C}$ in any combination; the only cubic object is the antisymmetric map $\Lambda^3 4 \simeq \bar{4}$, which is $\mathfrak{su}(4)$- but not $\mathfrak{u}(4)$-covariant and

exists only for $N = 4$. The super-Jacobi identity reduces the mixed bracket to $q = -p$; the $\mathfrak{sl}(4)$ (traceless) requirement then gives $p = q = 0$.

Scope of the statements

Everything above concerns *this* graded space with *these* bracket conventions. A different real-form convention can change the intermediate one-parameter family (even to $K = 0$), although the $\mathfrak{sl}(4)$ requirement always annihilates the bracket. These are not no-go theorems for arbitrary discrete models. They are five checkable criteria—positivity, gauge content, chirality, normalisation, embedding—for constructions of this type.

9 Known versus new

Item	Status	Basis
2D area law $\sigma_\infty = -\ln(I_1/I_0)$, confinement at all β	known	Savit 1980
$Z_F = \sum_n I_n^F$ on a closed surface	known for torus; extended here	character expansion
F-universality, genus independence	new theorem (elementary)	Result 2
Exact loop law $\sum_n I_n^A I_{n+1}^{F-A}/Z$	new theorem, to our knowledge	Result 3
Two screening channels at $A = F/2$	new	Eq. (8)
Asymptotic expansion with remainder bound	new theorem	Eqs. (12)–(14)
$\sigma_{\text{oct}} = 0.800988$ vs $\sigma_\infty = 0.806562$	new benchmark	Table 2
Villain closed form	known	Villain 1975
$\beta_* = F/2\pi$ (extensive self-dual coupling)	new theorem (elementary)	Result 4
Wilson–Villain dictionary ambiguity	new discussion	Sec. 4.3
$\lvert\det(m + D)\rvert^2$ and four flat zero modes	model-specific new results	Results 5–6
Algebra census, Jacobi locus, chirality	model-specific new audit	Sec. 6
$\pi\sqrt{3}\rho^4$ chance-level	new negative study	Appendix C

Table 1: Status of the results, separating known 2D $U(1)$ theory, new proven statements, new numerical benchmarks, and model-specific interpretations. "New" means not found in the standard references cited; the elementary items are new statements with short proofs, not new techniques.

10 Code, dependencies, verification

Path	Content
`octahedron_NPB.tex/.pdf`	manuscript (elsarticle, 28 pages)
`cover_letter_NPB.*`	cover letter
`guide_en.pdf`	this guide
`code/`	13 runnable scripts
`figures/`	paper figures (PDF + PNG)
`guide_figures/`	guide figures, including the graphical abstract
`logs/`	per-script logs and JSON outputs
`TEST_REPORT.md`	test report (28/28 assertions)
`CODE_GUIDE.md`	per-script description and paper mapping

```
cd NPB_octahedron
python3 code/verify_all.py   # 28/28 checks
bash run_all_tests.sh      # all scripts + logs
```

Dependencies: Python 3.12, numpy 1.26, scipy 1.11, mpmath 1.2, matplotlib 3.6; LaTeX with `elsarticle`. Closed-form sums use 40–60 digits; Monte Carlo is used only as an independent cross-check of the loop formula.

Script	Purpose	Paper
`exact_gauge.py`	Z_F, σ_F, loops, specific heat	Tables 2–4, Figs. 1–2
`finite_size_asymptotic.py`	expansion and error control	Eqs. (12)–(14)
`wilson_loops.py`	exact loops vs Monte Carlo	Sec. 3.1
`villain_general.py`	Villain solution, duality, dictionaries	Sec. 4
`triangulation_uniqueness.py`	uniqueness of the 6-vertex triangulation	Lemma 1
`dirac_operator_checks.py`	D_0 vs $D(m)$, conjugation, grading, gap	Sec. 5
`fermion_exact.py`, `fermion_scan.py`	exact matter averages	Eq. (29)
`fermion_spectrum.py`	Γ spectrum and kernel	Result 6
`algebra_census.py`	invariant tensor census	Result 7
`verify_jacobi.py`	mixed-bracket Jacobi	Result 8
`verify_all.py`	28 assertions, the gatekeeper	all
`make_figures.py`, `make_guide_figures.py`	paper and guide figures	—

Headline numbers (all asserted)

Quantity	Value
$Z(1) = \sum_n I_n(1)^8$	6.6224348046478
$\langle e^{i\theta_f}\rangle_{\rm oct}(1)$	0.4488851447705
$\sigma_{\rm oct}(1)$; $\sigma_\infty(1)$	0.8009882263; 0.8065623460
$\langle W_A\rangle$, $A = 1, 2, 3, 4$	0.4488851, 0.2066158, 0.1063616, 0.0791731
$c(1)$; peak	2.932656; 5.745 at $\beta \simeq 2.05$
Villain β_*	$F/2\pi$ (1.27324 for $F = 8$)
$\langle \det D_0\rangle$; $\langle (\det D_0)^2\rangle$	4.257698; 53.180808
Γ spectrum	$\{0, 0, 0, 0, \pm 2\sqrt{3}i\}$, rank 2

11 Frequently asked questions

"A finite volume is analytic—why publish it?" Because this is not a trivial free theory: it gives a closed-form benchmark for the infinite-volume confining string tension, and the finite-size correction is exponentially small with an explicit bound. Anyone computing on a small torus can check against it.

"Did the $1/F$ claim change?" Yes. The earlier draft quoted a power law; the exact evaluation gives $r^{F-2}[1 + \cdots]$, i.e. an exponential with an analytic prefactor. The present version includes the derivation, the range of validity, and a table of exact versus asymptotic values with relative errors.

"Is this a no-go for all discrete models?" No. Every "cannot" is scoped to this complex, this minimal Dirac operator, and this graded space with these bracket conventions. The paper states five criteria that other constructions can confront one by one.

"Does the code actually run?" Yes: `verify_all.py` passes 28/28 assertions, every script has its own log, closed forms use 40–60 digits, and the loop formula is cross-checked by an independent Monte Carlo. The code is archived at `https://github.com/csoftxyz/RIA_EISA/tree/main/08/code_and_logs`.

This guide, the manuscript, the code and the logs form one submission package. AI assistance: language polishing and code testing/verification were assisted by the DeepSeek AI assistant; all original formulas and derivations are the authors' own work.